

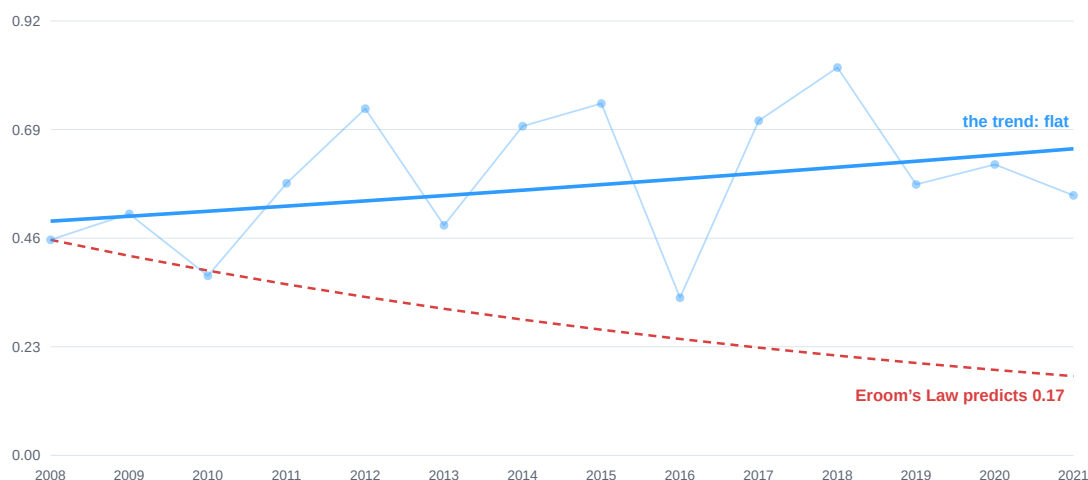
Pharma's most quoted number stopped being true

The US, 2008–2021: Eroom's Law said drug productivity halves every nine years

US new molecular entities approved per billion dollars of US pharmaceutical business R&D, in constant prices.

Red dashed: where the ratio should be if the law still held. Blue: the trend through what actually happened, with the yearly figures behind it.

It did not continue: the ratio is flat to rising, and the law's own rate is rejected in every specification tested.



By 2021 the law implies 0.17 approvals per billion. The figure was 0.55 — more than 3 times higher.

The swings are approvals, not spending: real R&D grows smoothly and accounts for only 10% of the ratio's movement.

This tests only whether the published rate continued. The paper's own 1950–2010 window cannot be re-measured from open data.

Data: FDA Drugs@FDA submission records (new molecular entities) and Eurostat rd_e_berdind2 NACE C21, deflated by the US GDP price index · namikakmandev.github.io/pharma-eroom.html

Scannell et al., Nature Reviews Drug Discovery, 2012

Moore's Law, backwards

Moore's Law says computing power doubles every two years. In 2012 four researchers went looking for the same thing in drug research and found its exact opposite.

New drugs approved per billion dollars of research had **halved every nine years since 1950**. Sixty years, no exceptions. They spelled Moore backwards and called it **Eroom's Law**.

Its data stops in 2010. It is still on slides today.

So I rebuilt it

Approvals from the FDA's own submission records — the count reproduces the FDA's published figures almost exactly. Spending from Eurostat's R&D accounts, deflated.

+0.021

fitted slope per year, 2008–2021

-0.077

what Eroom's Law implies

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US specifications reject it

By 2021 the law implies **0.17** approvals per billion dollars. The figure was **0.55** — more than 3 times higher.

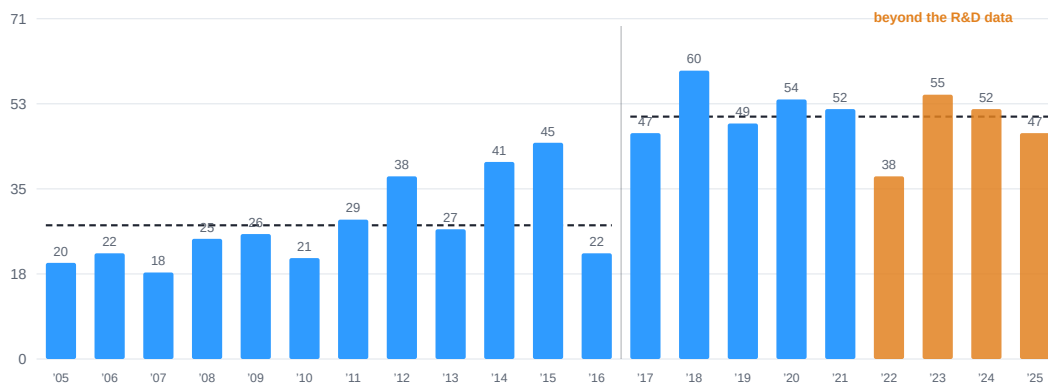
This tests whether the published rate continued. The paper's own 1950–2010 window cannot be re-measured: no open R&D series reaches back that far.

Was it AI?

The rise happened before AI could have caused it

New molecular entities approved by the FDA each year. The step up is complete by 2017; AI-designed molecules reached the clinic around 2020 and have almost no approvals yet. This does not test what did cause it.

2005–2016: 28 a year on average. 2017–2025: 50. **1.8× more.**



The ratio chart stops in 2021 because the R&D series does. Approvals run to 2025; 2026 is a part year at the data cut-off and is left out.

The orange bars are years with approvals but no matching R&D figure, so they cannot enter the productivity ratio.

Data: FDA Drugs@FDA submission records, original applications approved as submission class 7 or 8 · namikakmandev.github.io/pharma-eroom.html

No — on timing alone

Approvals step up once, and the step is complete by **2017**: 28 a year before, 50 after. The break year is chosen by fit, not by eye.

AI-designed molecules reached the clinic around 2020 and have almost no approvals yet. **A cause cannot postdate its effect.**

What did cause it is not tested here and this deck does not guess. The step is established; the cause stays open.

The honest catch

The ratio held up because **approvals rose**, not because spending fell. Real R&D grew steadily throughout and accounts for only **10%** of the ratio's movement.

So this is an approvals story with a slow denominator. Calling it a productivity story would overstate it.

Date the claim

If the nine-year halving goes into a business case, say which years it describes: **1950–2010**. On the years since, it is rejected.

Method, every number and both scripts:

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Data: FDA Drugs@FDA submission records; Eurostat rd_e_berdindr2 NACE C21; US GDP deflator.